

EE582 Homework 01 Report

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1 Question 1

A shunt harmonic filter (series RLC) is shown in Fig. 1(a). It is designed for a three-phase 13.8 kV, 60 Hz system to shunt the 7th harmonic current. The circuit parameters are:

- $r_L = 1 \Omega$
- $L = 19 \text{ mH}$
- $C = 8.2 \mu\text{F}$

Step 1: Write the KVL equation:

$$V_{\text{dc}} - r_L i_L - L \frac{di_L}{dt} - v_C = 0$$

Or equivalently:

$$L \frac{di_L}{dt} + r_L i_L + v_C = V_{\text{dc}}$$

With $i_L = C \frac{dv_C}{dt}$ and $v_C = \frac{1}{C} \int i_L dt$.

Step 2: Standard second-order ODE:

$$\frac{d^2 i_L}{dt^2} + \frac{r_L}{L} \frac{di_L}{dt} + \frac{1}{LC} i_L = 0$$

Step 3: Characteristic equation:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

Step 4: Calculate parameters:

$$\begin{aligned}\omega_n &= \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{19 \times 10^{-3} \times 8.2 \times 10^{-6}}} = 2533.5 \text{ rad/s} \\ \zeta &= \frac{r_L}{2\sqrt{\frac{L}{C}}} = \frac{1}{2\sqrt{\frac{19 \times 10^{-3}}{8.2 \times 10^{-6}}}} = 0.0104 \\ \omega_d &= \omega_n \sqrt{1 - \zeta^2} = 2533.3 \text{ rad/s}\end{aligned}$$

Step 5: Quality factor:

$$Q_{\text{factor}} = \frac{\omega_n L}{r_L} = \frac{2533.5 \times 0.019}{1} = 48.136$$

Step 6: Eigenvalues:

$$\lambda = -\zeta\omega_n \pm j\omega_d = -26.316 \pm 2533.3j$$

Step 7: Reactive power:

$$Q_{\text{kVAr}} = -602.04 \text{ kVAr}$$

All calculations confirm the system is very underdamped ($\zeta = 0.0104$).

Summary of Key Results:

- Natural frequency: $\omega_n = 2533.5 \text{ rad/s}$
- Damping ratio: $\zeta = 0.0104$
- Damped frequency: $\omega_d = 2533.3 \text{ rad/s}$
- Quality factor: $Q = 48.136$
- Eigenvalues: $-26.316 \pm 2533.3j$
- Reactive power: -602.04 kVAr

2 Question 2

Consider the series RLC circuit with parameters r_L , L , and C as in Q1. The equations are:

KVL:

$$V_{dc} - r_L i_L - L \frac{di_L}{dt} - v_C = 0 \quad (1)$$

where i_L is the inductor current and v_C is the capacitor voltage.

Capacitor current:

$$i_L = C \frac{dv_C}{dt} \quad (2)$$

Substitute (2) into (1):

$$V_{dc} - r_L C \frac{dv_C}{dt} - LC \frac{d^2 v_C}{dt^2} - v_C = 0$$

Rearrange to standard form:

$$LC \frac{d^2 v_C}{dt^2} + r_L C \frac{dv_C}{dt} + v_C = V_{dc}$$

Divide both sides by LC :

$$\frac{d^2 v_C}{dt^2} + \frac{r_L}{L} \frac{dv_C}{dt} + \frac{1}{LC} v_C = \frac{V_{dc}}{LC}$$

This is a standard second-order nonhomogeneous ODE. The homogeneous solution is:

$$\frac{d^2 v_C}{dt^2} + \frac{r_L}{L} \frac{dv_C}{dt} + \frac{1}{LC} v_C = 0$$

Eigenvalue computation: The characteristic equation is:

$$s^2 + \frac{r_L}{L} s + \frac{1}{LC} = 0$$

The eigenvalues are:

$$s = \frac{-\frac{r_L}{L} \pm \sqrt{\left(\frac{r_L}{L}\right)^2 - 4 \cdot \frac{1}{LC}}}{2}$$

For the given values ($r_L = 1 \, \Omega$, $L = 19 \, \text{mH}$, $C = 8.2 \, \mu\text{F}$):

$$\begin{aligned} \frac{r_L}{L} &= \frac{1}{0.019} = 52.63 \, \text{s}^{-1} \\ \frac{1}{LC} &= \frac{1}{0.019 \times 8.2 \times 10^{-6}} = 6,430,041 \, \text{s}^{-2} \end{aligned}$$

Plug into the quadratic formula:

$$s = \frac{-52.63 \pm \sqrt{(52.63)^2 - 4 \times 6,430,041}}{2}$$

The discriminant is negative, so the roots are complex conjugates:

$$s = -26.316 \pm j 2533.3$$

Interpretation: The eigenvalues are complex conjugates with negative real parts, confirming the circuit is underdamped. Based on the very low value of the damping ratio $\zeta = 0.0104$ calculated in Q1, the transient response will exhibit oscillatory behavior with a slow decay.

General solution:

$$v_C(t) = v_{C,\infty} + [v_{C,0} - v_{C,\infty}] \left[1 - \frac{1}{\sqrt{1 - \zeta^2}} e^{-\zeta \omega_n t} \sin(\omega_d t + \phi) \right]$$

where $v_{C,\infty}$ is the steady-state value, $v_{C,0}$ is the initial value, and $\omega_d = \omega_n \sqrt{1 - \zeta^2}$.

For a step input, $v_{C,\infty} = x_{SS}$, and the solution simplifies to:

$$x(t) = x_{SS} \left[1 - \frac{1}{\sqrt{1 - \zeta^2}} e^{-\zeta \omega_n t} \sin(\omega_d t + \phi) \right]$$

For the given values ($r_L = 1 \, \Omega$, $L = 19 \, \text{mH}$, $C = 8.2 \, \mu\text{F}$), the numeric solution is:

$$x(t)[kV] = 11.3 * [1 - 1.00005 e^{-26.34 t} \sin(2533.3 t + 1.5604)]$$

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